Lecture 5

Differentiability

Mathematically, on a curve C with equation $y = f(x)$, the gradient of a curve at a point $(x, f(x_0)$ is defined as the rate at which the value of y changes as x is changed from x_0 to another value infinitesimally close (infinitely close) to x_0.

i.e., (x, y) changes from $(x_0, f(x_0))$ to $(x_0 + h, f(x_0 + h))$ where $h \to 0$.

So, according to definition, gradient of the curve at $(x, f(x_0))$ is given by,

$$m = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0}$$

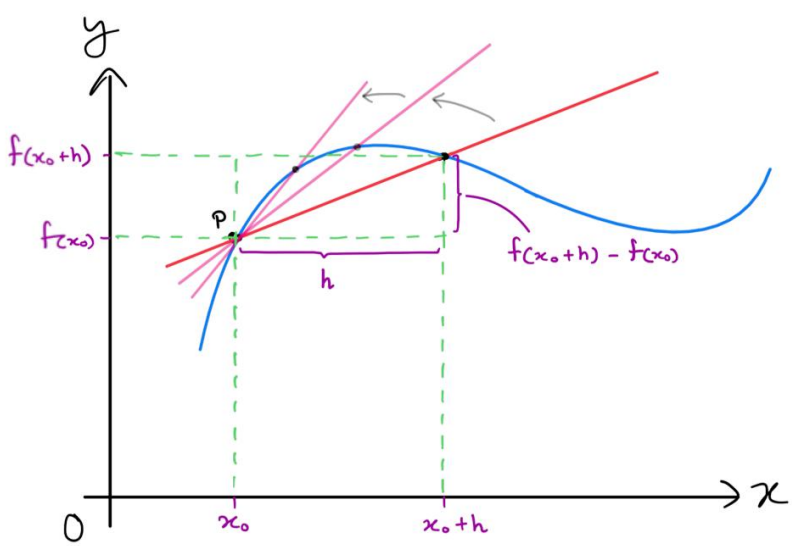
$$= \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Recall that the tangent line to a curve C at point $P(x_0, f(x_0))$ is just a secant line through P and another point on C infinitesimally close to P.

So, gradient of the tangent $= \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{(x_0+h) - x_0}$

$$= \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$= \text{gradient of C at } (x_0, f(x_0))$$



So, if the limit, $\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}$ exists, then it can be interpreted either as the slope of the tangent line to the curve $y = f(x)$ at $x = x_0$, or as the instantaneous rate of change of y with respect to x at $x = x_0$.

<u>Derivative:</u> The derivative of a function $f(x)$, denoted by $f'(x)$, is

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

It is called the derivative of f with respect to x.

If $y = f(x)$, then $\frac{dy}{dx} = \frac{d}{dx}(f(x))$ = $D_x[f(x)]$

$\Rightarrow y' = f'(x)$

<u>Derivative of a function $f(x)$ at point $x = a$:</u>

Let the function $f(x)$ be defined at $x = a$, then $f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ is called the derivative of $f(x)$ at $x = a$, provided the limit exists (L.H.D = R.H.D).

$$\frac{dy}{dx}|_{x=a} = f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

<u>Differentiable Function:</u> A function is said to be differentiable at $x = a$ if the derivative exists at $x = a$.

i. $f(x)$ must be continuous.

ii. L.H.D = R.H.D

<u>Left-hand derivative (LHD):</u>

If $\lim_{h\to 0^-}\frac{f(a+h)-f(a)}{h}$ exists, then it is called the left-hand derivative of $f(x)$ at $x = a$, denoted by

$$Lf'(a) = \lim_{h\to 0^-}\frac{f(a+h)-f(a)}{h}$$

<u>Right-hand derivative (RHD):</u>

If $\lim_{h\to 0^+}\frac{f(a+h)-f(a)}{h}$ exists, then it is called the right-hand derivative of $f(x)$ at $x = a$, denoted by

$$Rf'(a) = \lim_{h\to 0^+}\frac{f(a+h)-f(a)}{h}$$

<u>Theorem:</u> If a function is differentiable at a point, then it is also continuous at that point.

Example: Find the derivative of $f(x) = \sin x$ using the limit definition.

Solution: Here, $f(x) = \sin x$. So, $f(x+h) = \sin(x+h)$.

$$\begin{aligned}
\text{Now, } f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h} \\
&= \lim_{h\to 0} \frac{2\cos\left(\frac{x+h+x}{2}\right)\sin\left(\frac{x+h-x}{2}\right)}{h} \qquad \left[\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)\right] \\
&= \lim_{h\to 0} \frac{2\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} \\
&= \lim_{h\to 0} 2\cos\left(x+\frac{h}{2}\right)\cdot\frac{\sin\left(\frac{h}{2}\right)}{h} \\
&= \lim_{h\to 0} 2\cos\left(x+\frac{h}{2}\right)\cdot\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\cdot\frac{1}{2} \\
&= 2\times cosx \times 1 \times \frac{1}{2} \qquad \left[\lim_{x\to 0}\frac{\sin ax}{ax} = 1\right] \\
&= \cos x
\end{aligned}$$

Hence, $f'(x) = \cos x$.

Example: Find the derivative of $f(x) = \cot x$ using the limit definition.

Solution:

Here, $f(x) = \cot x$. So, $f(x+h) = \cot(x+h)$.

$$\begin{aligned}
\text{Now, } f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{\cot(x+h) - \cot x}{h} \\
&= \lim_{h\to 0} \frac{\frac{\cos(x+h)}{\sin(x+h)} - \frac{\cos x}{\sin x}}{h} \\
&= \lim_{h\to 0} \frac{\frac{\sin x\cos(x+h) - \cos x\sin(x+h)}{\sin(x+h)\sin x}}{h} \\
&= \lim_{h\to 0} \frac{\sin(x-x-h)}{h\sin(x+h)\sin x} \qquad [\sin A\cos B - \cos A\sin B = \sin(A-B)]
\end{aligned}$$

$$= \lim_{h\to 0} \frac{\sin(-h)}{h\sin(x+h)\sin x}$$

$$= \lim_{h\to 0} \frac{-\sin h}{h\sin(x+h)\sin x} \qquad [\sin(-x) = -\sin x]$$

$$= \lim_{h\to 0} \frac{-\sin h}{h} \times \lim_{h\to 0} \frac{1}{\sin(x+h)\sin x}$$

$$= -1 \times \frac{1}{\sin x \times \sin x}$$

$$= -\frac{1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x$$

Hence, $f'(x) = -\operatorname{cosec}^2 x$.

H.W. Find the derivatives of the following functions using limit definition: $f(x) = \tan x$, $f(x) = \cos x$, $f(x) = \sec x$, $f(x) = \operatorname{cosec} x$.

Example: $f(x) = \begin{cases} x^2 + 1 & , \quad x \le 1 \\ x & , \quad x > 1 \end{cases}$

Test differentiability at $x = 1$ and sketch the graph.

Solution: Here, $f(1) = (1)^2 + 1 = 2$

L.H.L. $= \lim_{x\to 1^-} f(x) \quad = \lim_{x\to 1^-} (x^2 + 1) = (1)^2 + 1 = 2$

R.H.L. $= \lim_{x\to 1^+} f(x) \quad = \lim_{x\to 1^+} x = 1$

As L.H.L. $\neq$ R.H.L., the limit doesn't exist and the function is not continuous at $x = 1$. So, $f(x)$ is not differentiable at $x = 1$.

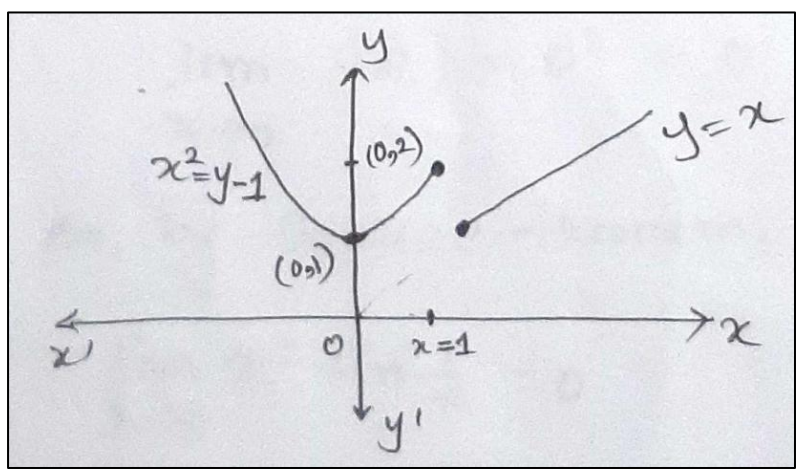
Sketch: If $x \le 1$, $f(x) = y = x^2 + 1$

$\Rightarrow y - 1 = x^2$

$\Rightarrow x^2 = y - 1$

If $x = 1$, $f(x) = y = (1)^2 + 1 = 2$

If $x > 1$, $f(x) = y = x$.



Example: $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & , \quad x \neq 0 \\ 0 & , \quad x = 0 \end{cases}$

Test differentiability at $x = 0$.

Solution: Here, $x = 0$. Thus, $f(0) = 0$.

Since, $\quad -1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

$$\Rightarrow -x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

Here, $\lim\limits_{x \to 0} (-x^2) = -0^2 = 0$

$$\lim_{x \to 0} (x^2) \quad = 0^2 = 0$$

So, by Squeeze Theorem,

$$\lim_{x \to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

As, $f(0) = \lim\limits_{x \to 0} f(x) = 0$, the function is continuous at $x = 0$.

For differentiability:

L.H.D. $= Lf'(0) = \lim\limits_{h \to 0^-} \frac{f(0+h) - f(0)}{h}$

$$= \lim_{h \to 0^-} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0^-} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h}$$

$$= \lim_{h \to 0^-} h \sin\left(\frac{1}{h}\right)$$

$$= 0 \qquad \text{[By using squeeze theorem, we know, } \lim_{h \to 0} h \sin\left(\frac{1}{h}\right) = 0\text{]}$$

Note: You cannot write

$\lim\limits_{h \to 0} h \sin\left(\frac{1}{h}\right) = \lim\limits_{h \to 0} \frac{\sin\left(\frac{1}{h}\right)}{\frac{1}{h}} = 1.$

It's wrong because $\frac{1}{h} \to \infty$, not 0.

R. H. D. $= Rf'(0) \quad = \lim\limits_{h \to 0^+} \frac{f(0+h) - f(0)}{h}$

$$= \lim_{h \to 0^+} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0^+} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h}$$

$$= \lim_{h \to 0^+} h \sin\left(\frac{1}{h}\right)$$

$$= 0 \qquad \text{[By using squeeze theorem, we know, } \lim_{h \to 0} h \sin\left(\frac{1}{h}\right) = 0\text{]}$$

Since $L.H.D. = R.H.D.$, $f(x)$ is differentiable at $x = 0$.

Example: $f(x) = \begin{cases} 5x - 4 & , \quad 0 < x \leq 1 \\ 4x^2 - 3x & , \quad 1 < x < 2 \\ 3x + 4 & , \quad x \geq 2 \end{cases}$

Find the continuity and differentiability at $x = 1,2$.

Solution:

<u>For $x = 1$:</u>

For continuity:

Here, $x = 1$

$f(1) = 5(1) - 4 = 1$

L.H.L. $= \lim_{x \to 1^-} f(x) \quad = \lim_{x \to 1^-} (5x - 4) = 5(1) - 4 = 1$

R.H.L. $= \lim_{x \to 1^+} f(x) \quad = \lim_{x \to 1^+} (4x^2 - 3x) = 4(1)^2 - 3(1) = 4 - 3 = 1$

Here, $f(1) = \lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = 1$

$\therefore f(x)$ is continuous at $x = 1$.

For differentiability:

$$\begin{aligned} \text{L. H. D.} = Lf'(1) \quad &= \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \to 0^-} \frac{[5(1+h) - 4] - [5(1) - 4]}{h} \\ &= \lim_{h \to 0^-} \frac{5 + 5h - 4 - 1}{h} \\ &= \lim_{h \to 0^-} \frac{5h}{h} \\ &= \lim_{h \to 0^-} 5 = 5 \end{aligned}$$

Note: As $h \to 0^-$, the value of h is negative here.So, $1 + h = 1 - ve < 1$. So, we will choose the first part of the function here where the interval is $0 < x \leq 1$.

$$\begin{aligned} \text{R. H. D.} = Rf'(1) \quad &= \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \to 0^+} \frac{4(1+h)^2 - 3(1+h) - [4 \times 1^2 - 3 \times 1]}{h} \\ &= \lim_{h \to 0^+} \frac{4(1 + 2h + h^2) - 3 - 3h - 1}{h} \\ &= \lim_{h \to 0^+} \frac{4 + 8h + 4h^2 - 4 - 3h}{h} \\ &= \lim_{h \to 0^+} \frac{4h^2 + 5h}{h} \\ &= \lim_{h \to 0^+} (4h + 5) = 4 \times 0 + 5 = 5 \end{aligned}$$

Note: As $h \to 0^+$, the value of h is positive here. So, $1 + h = 1 + ve > 1$. So, we will choose the second part of the function here where the interval is $1 < x < 2$.

H.W. Do the above math for $x = 2$.

H.W. $f(x) = x^2 + 2$. Is $f(x)$ differentiable at $x = 1$?

Example: $f(x) = \begin{cases} x^2, & x \le 1 \\ \sqrt{x}, & x > 1 \end{cases}$

Test differentiability at $x = 1$.

Solution: Here, $x = 1$. So, $f(1) = 1^2 = 1$

L.H.L. $= \lim_{x \to 1^-} f(x) \quad = \lim_{x \to 1^-} x^2 = 1^2 = 1$

R.H.L. $= \lim_{x \to 1^+} f(x) \quad = \lim_{x \to 1^+} \sqrt{x} = \sqrt{1} = 1$

Here, $f(1) = \lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = 1$

$\therefore f(x)$ is continuous at $x = 1$.

For differentiability:

$$\begin{aligned}
\text{L. H. D.} = Lf'(1) \quad &= \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0^-} \frac{(1+h)^2 - 1}{h} \\
&= \lim_{h \to 0^-} \frac{1 + 2h + h^2 - 1}{h} \\
&= \lim_{h \to 0^-} \frac{h^2 + 2h}{h} \\
&= \lim_{h \to 0^-} (h + 2) \\
&= 0 + 2 = 2
\end{aligned}$$

$$\begin{aligned}
\text{R. H. D.} = Rf'(1) \quad &= \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0^+} \frac{\sqrt{1+h} - 1^2}{h} \\
&= \lim_{h \to 0^+} \frac{\sqrt{1+h} - 1}{h} \qquad [\tfrac{0}{0} \text{ form}] \\
&= \lim_{h \to 0^+} \frac{\frac{1}{2\sqrt{1+h}} - 0}{1} \qquad [\text{as } \tfrac{d}{dx}(\sqrt{x}) = \tfrac{1}{2\sqrt{x}}] \\
&= \lim_{h \to 0^+} \frac{1}{2\sqrt{1+h}} = \frac{1}{2\sqrt{1+0}} = \frac{1}{2}
\end{aligned}$$

So, $Lf'(1) \neq Rf'(1)$,

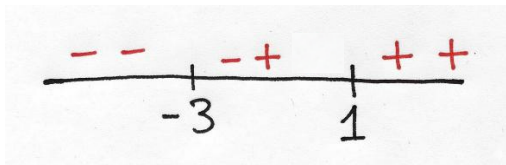
Hence, the function is not differentiable at $x = 1$.

H.W.

Example: $f(x) = |x-1| + |x+3|$

Discuss the differentiability at $x = 1$ and $x = -3$.

Solution:



If $x \geq 1$, let $x = 2$,

Then, $(x-1) = (2-1) = 1\ (+ve)$

and $(x+3) = (2+3) = 5\ (+ve)$

If $-3 \leq x < 1$, let $x = -2$.

Then, $(x-1) = (-2-1) = -3\ (-ve)$

and $(x+3) = (-2+3) = 1\ (+ve)$

If $x < -3$, let $x = -4$.

Then, $(x-1) = (-4-1) = -5\ (-ve)$

and $(x+3) = (-4+3) = -1\ (-ve)$

So, $f(x) = \begin{cases} (x-1)+(x+3) & , \quad x \geq 1 \\ -(x-1)+(x+3), & -3 \leq x < 1 \\ -(x-1)-(x+3), & x < -3 \end{cases}$

$\therefore f(x) = \begin{cases} 2x+2 & , \quad x \geq 1 \\ 4 & , \quad -3 \leq x < 1 \\ -2x-2 & , \quad x < -3 \end{cases}$

<u>For $x = 1$</u>

For continuity:

If $x = 1$, then, $f(x) = 2x + 2$

So, $f(1) = 2 \times 1 + 2 = 4$

Now, L.H.L. $= \lim_{x \to 1^-} f(x) \quad = \lim_{x \to 1^-} 4 = 4$

R.H.L. $= \lim_{x \to 1^+} f(x) \quad = \lim_{x \to 1^+} (2x + 2) = 2 \times 1 + 2 = 4$

As $f(1) = L.H.L. = R.H.L. = 4, f(x)$ is continuous at $x = 1$.

For differentiability:

$$\begin{aligned}
\text{L. H. D.} = Lf'(1) \quad &= \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0^-} \frac{4 - 4}{h} \\
&= \lim_{h \to 0^-} \frac{0}{h} \\
&= \lim_{h \to 0^-} 0 \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\text{R. H. D.} = Rf'(1) \quad &= \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0^+} \frac{2(1+h) + 2 - (2 \times 1 + 2)}{h} \\
&= \lim_{h \to 0^+} \frac{4 + 2h - 4}{h} \\
&= \lim_{h \to 0^+} \frac{2h}{h} \\
&= \lim_{h \to 0^+} 2 \\
&= 2
\end{aligned}$$

As L.H.D. ≠ R.H.D., $f(x)$ is not differentiable at $x = 1$.

<u>For $x = -3$</u>

For continuity:

If $x = -3$, then, $f(x) = 4$

So, $f(-3) = 4$

Now, L.H.L. $= \lim_{x \to -3^-} f(x) \quad = \lim_{x \to -3^-} (-2x - 2) = -2(-3) - 2 = 4$

R.H.L. $= \lim_{x \to -3^+} f(x) \quad = \lim_{x \to -3^+} 4 = 4$

As $f(-3) = L.H.L. = R.H.L. = 4, f(x)$ is continuous at $x = -3$.

For differentiability:

$$\begin{aligned}
\text{L. H. D.} = Lf'(-3) \quad &= \lim_{h\to 0^-} \frac{f(-3+h)-f(-3)}{h} \\
&= \lim_{h\to 0^-} \frac{-2(-3+h)-(-2(-3)-2)}{h} \\
&= \lim_{h\to 0^-} \frac{6-2h-2-4}{h} \\
&= \lim_{h\to 0^-} \frac{4-2h-4}{h} \\
&= \lim_{h\to 0^-} \frac{-2h}{h} \\
&= \lim_{h\to 0^-} (-2) \\
&= -2
\end{aligned}$$

$$\begin{aligned}
\text{R. H. D.} = Rf'(-3) \quad &= \lim_{h\to 0^+} \frac{f(-3+h)-f(-3)}{h} \\
&= \lim_{h\to 0^+} \frac{4-4}{h} \\
&= \lim_{h\to 0^+} \frac{0}{h} \\
&= \lim_{h\to 0^+} 0 \\
&= 0
\end{aligned}$$

As L.H.D ≠ R.H.D, $f(x)$ is not differentiable at $x = -3$.

Example: $f(x) = \begin{cases} 3+2x, & -\frac{3}{2} < x \leq 0 \\ 3-2x, & 0 < x < \frac{3}{2} \\ -3-2x, & x \geq \frac{3}{2} \end{cases}$

Discuss the differentiability at $x = 0$ and $x = \frac{3}{2}$.

Solution:

<u>For $x = 0$</u>

For continuity:

If $x = 0$, then, $f(x) = 3 + 2x$

So, $f(0) = 3 + 2 \times 0 = 3$

Now, L.H.L. $= \lim_{x\to 0^-} f(x) \quad = \lim_{x\to 0^-} (3+2x) = 3 + 2 \times 0 = 3$

R.H.L. $= \lim_{x\to 0^+} f(x) \quad = \lim_{x\to 0^+} (3-2x) = 3 - 2 \times 0 = 3$

As $f(0) = L.H.L. = R.H.L. = 3$, $f(x)$ is continuous at $x = 0$.

For differentiability:

$$\begin{aligned}
\text{L. H. D.} = Lf'(0) \quad &= \lim_{h\to 0^-} \frac{f(0+h)-f(0)}{h} \\
&= \lim_{h\to 0^-} \frac{f(h)-f(0)}{h} \\
&= \lim_{h\to 0^-} \frac{3+2h-3}{h} \\
&= \lim_{h\to 0^-} \frac{2h}{h} \\
&= \lim_{h\to 0^-} 2 \\
&= 2
\end{aligned}$$

$$\begin{aligned}
\text{R. H. D.} = Rf'(0) \quad &= \lim_{h\to 0^+} \frac{f(0+h)-f(0)}{h} \\
&= \lim_{h\to 0^+} \frac{f(h)-f(0)}{h} \\
&= \lim_{h\to 0^+} \frac{3-2h-3}{h} \\
&= \lim_{h\to 0^+} \left(-\frac{2h}{h}\right) \\
&= \lim_{h\to 0^+} (-2) \\
&= -2
\end{aligned}$$

As L.H.D. ≠ R.H.D., $f(x)$ is not differentiable at $x = 0$.

For $x = \frac{3}{2}$

For continuity:

If $x = \frac{3}{2}$, then, $f(x) = -3 - 2x$

So, $f\left(\frac{3}{2}\right) = -3 - 2 \times \frac{3}{2} = -6$

Now, L.H.L. $= \lim_{x\to\frac{3}{2}^-} f(x) \quad = \lim_{x\to\frac{3}{2}^-} (3-2x) = \left(3 - 2 \times \frac{3}{2}\right) = 0$

R.H.L. $= \lim_{x\to\frac{3}{2}^+} f(x) \quad = \lim_{x\to\frac{3}{2}^+} (-3-2x) = -3 - 2 \times \frac{3}{2} = -6$

As $L.H.L. \neq R.H.L.$, so the limit does not exist and the function $f(x)$ is not continuous at $x = \frac{3}{2}$.

So, $f(x)$ is not differentiable at $x = \frac{3}{2}$.

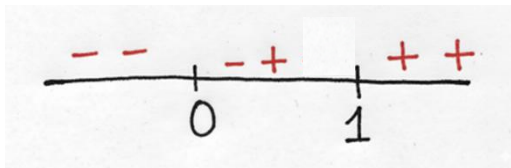
H.W $f(x) = \begin{cases} x^2 + 1 & , \quad x \leq 0 \\ x & , \quad 0 < x < 1 \\ \frac{1}{x} & , \quad x \geq 1 \end{cases}$

Discuss the differentiability at $x = 0$ and $x = 1$.

Example: $f(x) = |x - 1| + |x|$

Discuss the continuity and differentiability at $x = 1$.

Solution:



If $x \geq 1$, let $x = 2$,

Then, $(x - 1) = (2 - 1) = 1\ (+ve)$

and $x = 2 = (+ve)$

If $0 \leq x < 1$, let $x = 0$,

Then, $(x - 1) = (0 - 1) = -1\ (-ve)$

and $x = 0\ (+ve)$

If $x < 0$, let $x = -2$,

Then, $(x - 1) = (-2 - 1) = -3\ (-ve)$

and $x = -2\ (-ve)$

So, $f(x) = \begin{cases} (x - 1) + x & , \quad x \geq 1 \\ -(x - 1) + x & , \quad 0 \leq x < 1 \\ -(x - 1) - x & , \quad x < 0 \end{cases}$

$\therefore f(x) = \begin{cases} 2x - 1 & , \quad x \geq 1 \\ 1 & , \quad 0 \leq x < 1 \\ -2x + 1 & , \quad x < 0 \end{cases}$

<u>For $x = 1$</u>

For continuity:

If $x = 1$, then, $f(x) = 2x - 1$

So, $f(1) = 2 \times 1 - 1 = 1$

Now, L.H.L. $= \lim_{x \to 1^-} f(x) \quad = \lim_{x \to 1^-} 1 = 1$

R.H.L. $= \lim_{x \to 1^+} f(x) \quad = \lim_{x \to 1^+} (2x - 1) = 2 \times 1 - 1 = 1$

As $f(1) = L.H.L. = R.H.L. = 1$, $f(x)$ is continuous at $x = 1$.

For differentiability:

$$\begin{aligned}
\text{L. H. D.} = Lf'(1) \quad &= \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0^-} \frac{1 - 1}{h} \\
&= \lim_{h \to 0^-} \frac{0}{h} \\
&= \lim_{h \to 0^-} 0 \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\text{R. H. D.} = Rf'(1) \quad &= \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \to 0^+} \frac{2(1+h) - 1 - (2 \times 1 - 1)}{h} \\
&= \lim_{h \to 0^+} \frac{2 + 2h - 1 - 1}{h} \\
&= \lim_{h \to 0^+} \frac{2 + 2h - 2}{h} \\
&= \lim_{h \to 0^+} \frac{2h}{h} \\
&= \lim_{h \to 0^+} 2 \\
&= 2
\end{aligned}$$

As L.H.D. ≠ R.H.D., $f(x)$ is not differentiable at $x = 1$.

<u>H.W.</u> Do the above math for $x = 0$.

Example: $f(x) = \begin{cases} 1+x & , \quad x \le 0 \\ x & , \quad 0 < x < 1 \\ 2-x & , \quad 1 \le x \le 2 \\ 2x - x^2 & , \quad x > 2 \end{cases}$

Discuss the continuity and differentiability at $x = 0,1,2$.

Solution:

<u>For $x = 2$</u>

For continuity:

If $x = 2$, then, $f(x) = 2 - x$

So, $f(2) = 2 - 2 = 0$

Now, L.H.L. $= \lim_{x \to 2^-} f(x) \quad = \lim_{x \to 2^-} (2 - x) = 2 - 2 = 0$

R.H.L. $= \lim_{x \to 2^+} f(x) \quad = \lim_{x \to 2^+} (2x - x^2) = 2 \times 2 - 2^2 = 0$

As $f(2) = L.H.L. = R.H.L. = 0, f(x)$ is continuous at $x = 2$.

For differentiability:

$$\begin{aligned} \text{L. H. D.} = Lf'(2) \quad &= \lim_{h \to 0^-} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \to 0^-} \frac{2 - (2+h) - (2-2)}{h} \\ &= \lim_{h \to 0^-} \frac{2 - 2 - h - 0}{h} \\ &= \lim_{h \to 0^-} \frac{-h}{h} \\ &= \lim_{h \to 0^-} (-1) \\ &= -1 \end{aligned}$$

$$\begin{aligned} \text{R. H. D.} = Rf'(2) \quad &= \lim_{h \to 0^+} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \to 0^+} \frac{2(2+h) - (2+h)^2 - (2 \times 2 - 2^2)}{h} \\ &= \lim_{h \to 0^+} \frac{4 + 2h - 4 - 4\text{h} - h^2 - 0}{h} \\ &= \lim_{h \to 0^+} \frac{-2h - h^2}{h} \\ &= \lim_{h \to 0^+} \frac{h(-2-h)}{h} \\ &= \lim_{h \to 0^+} (-2 - \text{h}) \\ &= -2 \end{aligned}$$

As L.H.D. $\neq$ R.H.D., $f(x)$ is not differentiable at $x = 2$.

H.W. Do the above math for $x = 0,1$.